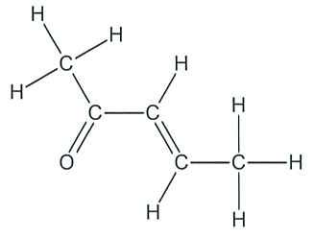
VCE Unit 4
SAC 4: ORGANIC COMPOUNDS AND SPECTROSCOPIC ANALYSIS

NAME: SOLUTIONS / MARKING SCHEME **Total Marks 56**

TEACHER (circle one): **MC** **SG** **RH** **SP** **NP**

Question 1

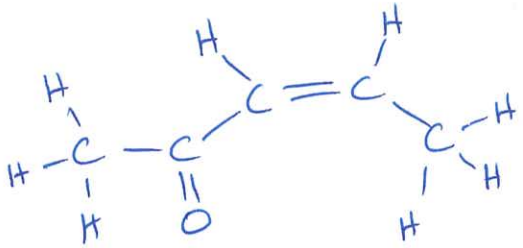
Carefully draw a full structure, ***showing all bonds***, that, with respect to the structure at the right, is



a) a structural isomer (1 mark)

- any valid structure with C_5H_8O that is different
- many possible answers

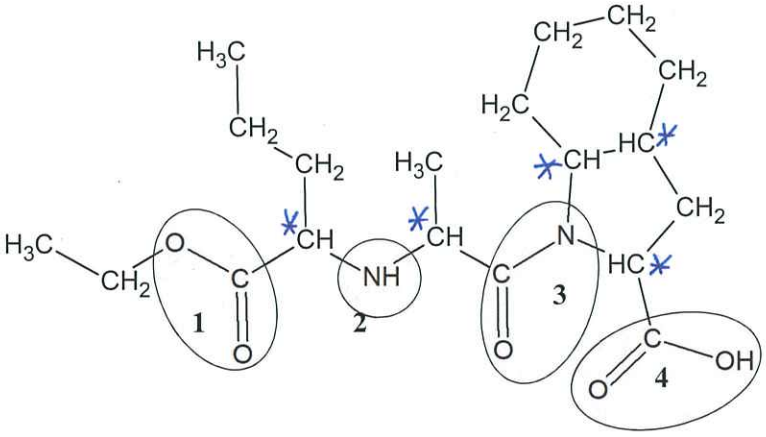
b) a stereoisomer (geometrical isomer) (1 mark)



- cis geometry clearly shown
- no other (structural) change.

Question 2

Below is the structure of perindopril, a drug frequently prescribed to treat high blood-pressure.



5 correct centres (2)
±1 " " (1)
otherwise (0)

a) ***Carefully*** label ***ALL*** of the *chiral centres* in this molecule with an asterisk (*). (2 marks)

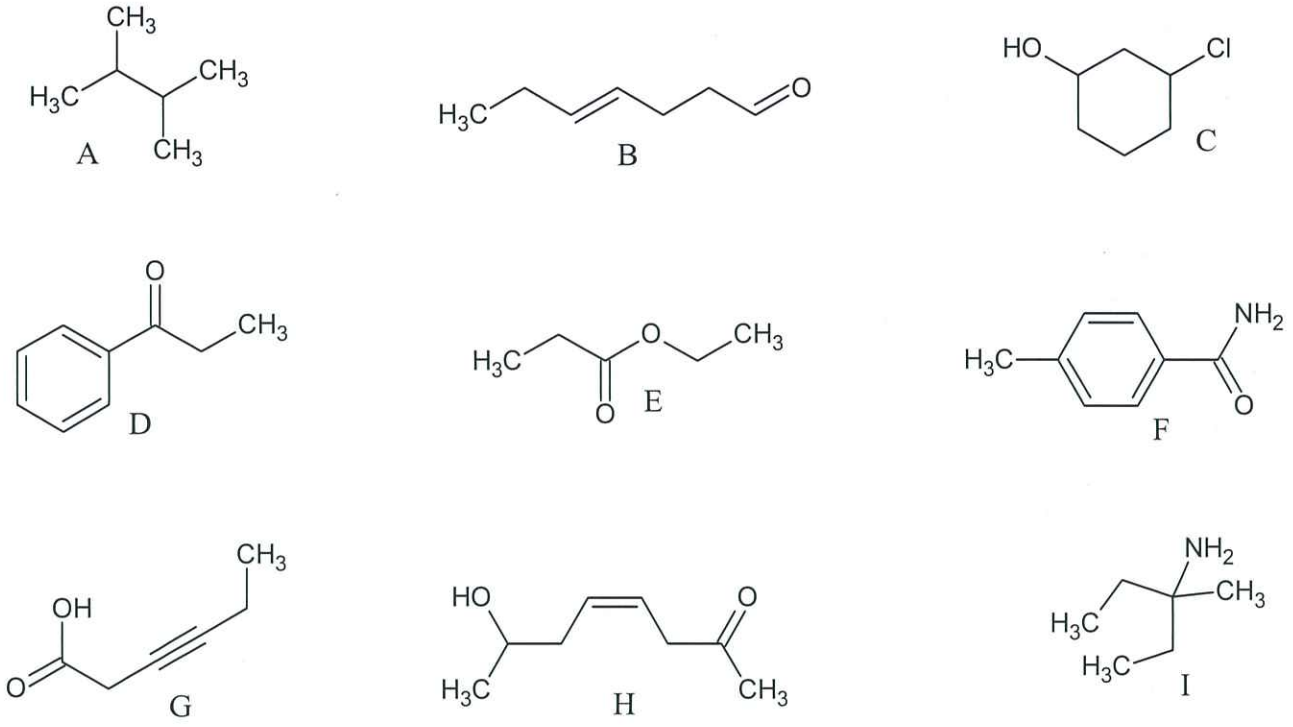
b) This molecule contains four of the functional groups we have studied. Identify each of the four different functional groups indicated. (4 marks)

1	Ester	2	Amine
3	Amide	4	Carboxy(l).

Distinction of 1°, 2°, 3° not required.

Question 3

Consider the following organic structures:

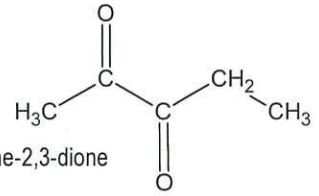


In each case below, circle the letters for ***ALL of the structures*** that match the description, or circle NONE if there is no structure that matches. Some structures/letters will be circled more than once. (11 marks)

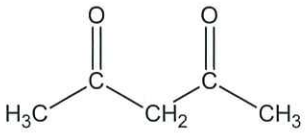
a) an ester	A	B	C	D	E	F	G	H	I	NONE
b) a trans- geometrical isomer	A	B	C	D	E	F	G	H	I	NONE
c) an aromatic compound	A	B	C	D	E	F	G	H	I	NONE
d) an alkyne	A	B	C	D	E	F	G	H	I	NONE
e) an alcohol (hydroxyl group)	A	B	C	D	E	F	G	H	I	NONE
f) a ketone	A	B	C	D	E	F	G	H	I	NONE
g) has at least one chiral centre	A	B	C	D	E	F	G	H	I	NONE
h) a carboxylic acid	A	B	C	D	E	F	G	H	I	NONE
i) an amine	A	B	C	D	E	F	G	H	I	NONE
j) a substituted cyclohexane	A	B	C	D	E	F	G	H	I	NONE
k) a structural isomer of hexane	A	B	C	D	E	F	G	H	I	NONE

Question 4

The chemical storeroom stocks the two isomers of pentanedione (di-ketones) shown below:



pentane-2,3-dione



pentane-2,4-dione

Unfortunately the labels have faded and because the two bottles were stored next to each other they have become indistinguishable. To identify them ^{1}H and ^{13}C NMR spectra were recorded for samples from each bottle.

Explain how each of the following features of these spectra would have been useful in distinguishing the two compounds. In each case justify your answer, including all relevant details.

a) The ***number*** of signals (ignoring chemical shift data) in the ^{13}C NMR spectrum of each compound. (2 marks)

pentane-2,3-dione - 5 (1) peaks. (5 C environments)

pentane-2,4-dione - 3 (1) peaks (3 C environments)

b) The ***number*** of low resolution signals (that is, ignoring any peak splitting and any chemical shift data) in the ^{1}H NMR spectrum of each compound, and the peak integrals (the number of H atoms contributing to each peak). (2 marks)

pentane-2,3-dione - 3 (½) signals in ratio 3:2:3 (½)

pentane-2,4-dione - 2 (½) signals in ratio 3:1 (½) (6:2)

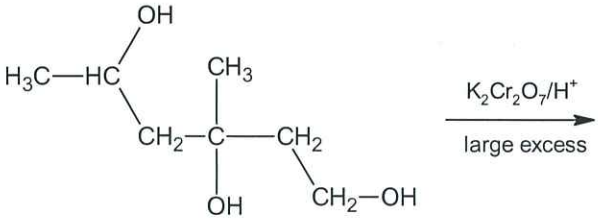
c) The splitting patterns observed in the high resolution ^{1}H NMR spectrum of each compound. (2 marks)

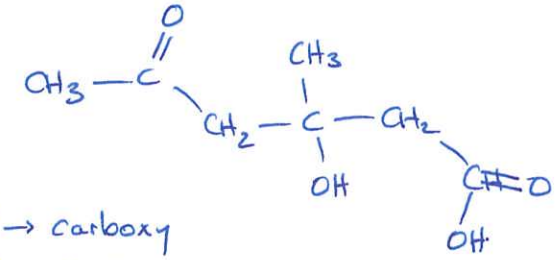
pentane-2,3-dione - singlet, quartet & triplet (1)
$-CH_2-CH_3$

pentane-2,4-dione - all singlets. (1)

Question 5

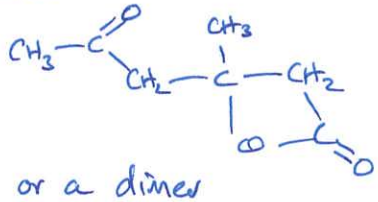
Draw the structure of the likely organic product formed when the compound shown below is heated with a large excess of acidified potassium dichromate until no further reaction is observed. (2 marks)





1° → carboxy
2° → ketone
3° → no change
(−1 for each incorrect)

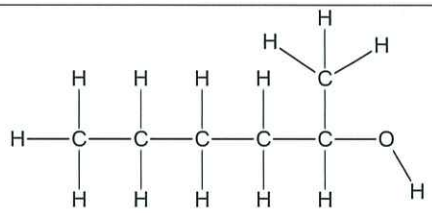
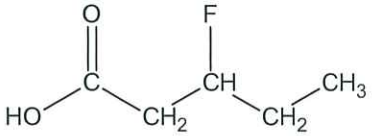
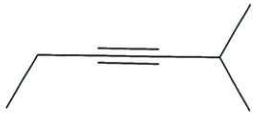
Also accept product of further reaction, eg



or a dimer

Question 6

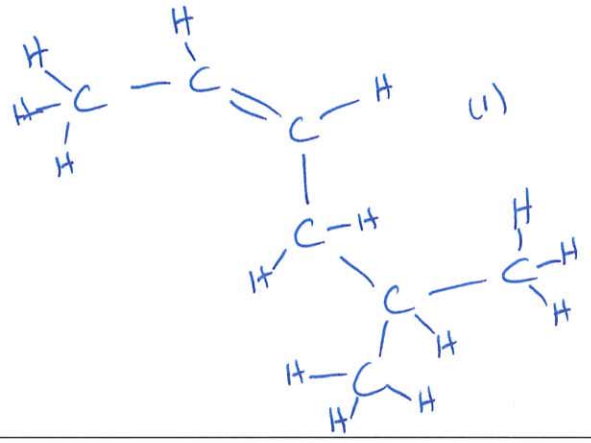
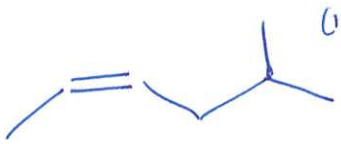
Give a systematic name for each of the structures shown below. (4 marks)

	hexan-2-ol.
	3-fluoropentanoic acid
	2-methylhex-3-yne
$CH_3(CH_2)_3COO(CH_2)_2CH_3$	propyl pentanoate

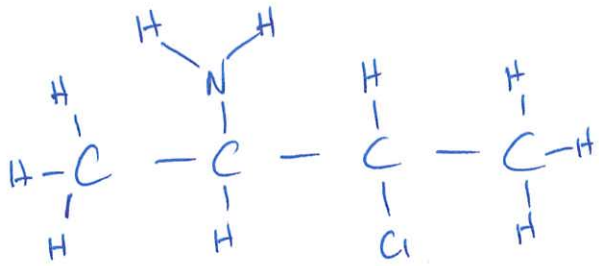
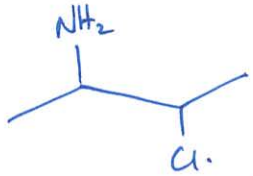
Question 7

For each compound named below draw both a complete structure, ***showing all atoms and bonds***, and also a skeletal structure.

a) *cis*-5-methylhex-2-ene (2 marks)

full structure	skeletal structure
 (1)	 (1)
⟷ consequential rule (1) if incorrect but consistent	

b) 3-chlorobutan-2-amine (2 marks)

full structure	skeletal structure
 (1)	 (1)
⟷ as above.	

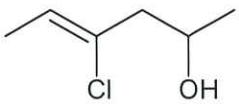
Question 8

Write a semi-structural formula for each of the following. (2 marks)

a) 5-aminoheptanoic acid

$CH_3CH_2CH(NH_2)CH_2CH_2CH_2COOH$ (1)

or minor variants eg $CH_3CH_2CH(NH_2)(CH_2)_3COOH$

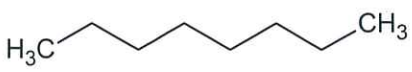
b) 

Brackets around OH and NH_2 are recommended but not required.

$CH_3CHClCH_2CH(OH)CH_3$ (1)

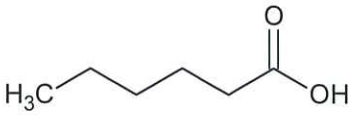
Question 9

Carefully account for the different boiling points observed for the three compounds below. (3 marks)



octane
M = 114 g mol^{-1}
bp = 126 °C

heptanal
M = 114 g mol^{-1}
bp = 153 °C



hexanoic acid
M = 116 g mol^{-1}
bp = 205 °C

(2) mks
−1 for each error/omission

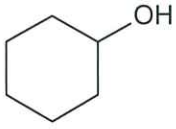
octane - dispersion forces only
heptanal - some dipole-dipole attractions
hexanoic acid - some hydrogen bonding

increasing strength of intermolecular bonding (1)

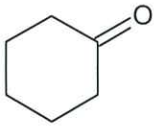
Question 10

Consider the following four structures.

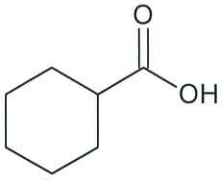
A



B

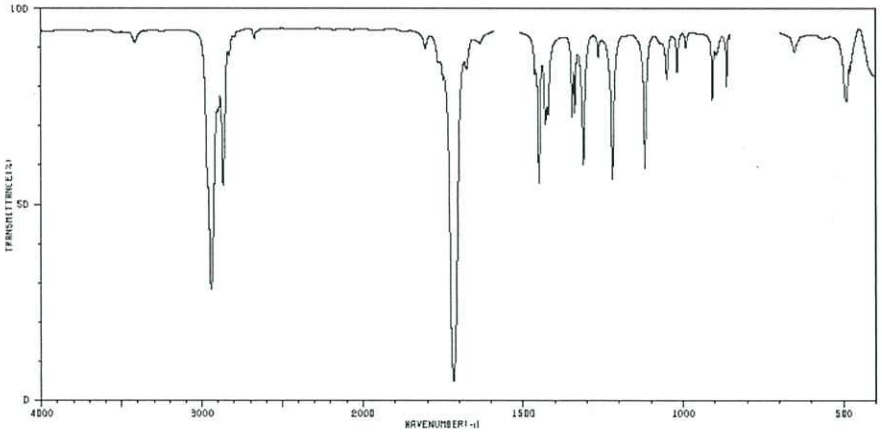


C



D

Which of these is most likely to have produced the following IR spectrum? Carefully justify your choice, by referring to all relevant conclusions that can be drawn from the spectrum. (3 marks)



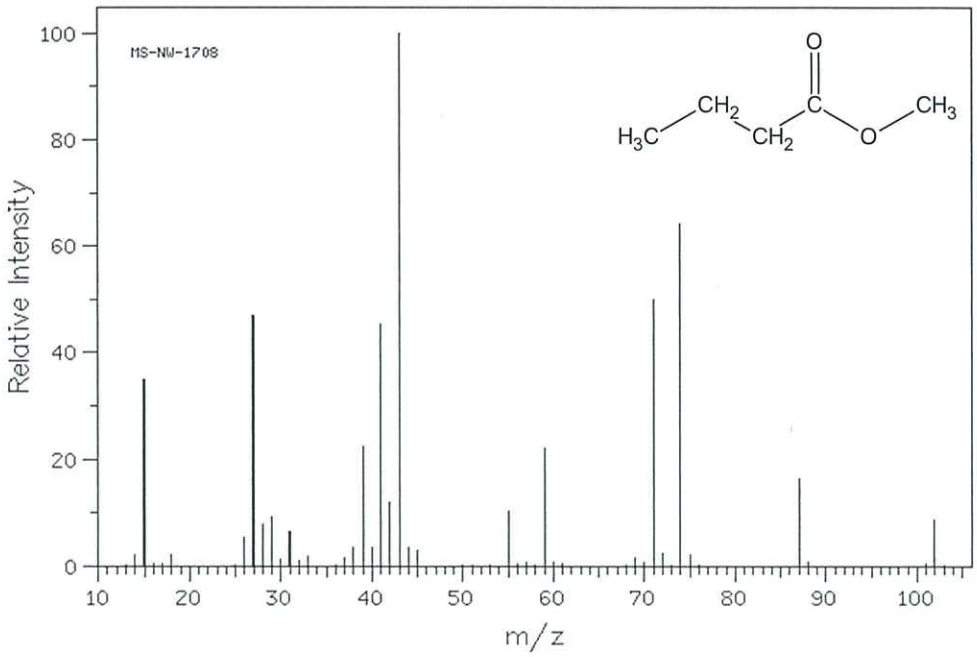
C (1)

- presence of strong signal (½) at ~1710 cm^{-1} ⇒ C=O group (½)
- absence of broad signal at > 3000 cm^{-1} (½) ⇒ no -OH or -COOH group. (½)

Question 11

At the right is a mass spectrum of methyl butanoate.

Refer to the spectrum and answer the following questions. (4 marks)



a) What are the m/z values for

i. the parent/molecular ion peak? ....102....

ii. the base peak? ....43....

b) The signals in the spectrum that are listed below can be attributed to fragments arising from simple fission (breaking) of one bond in the parent ion. Write the formula of the particle likely to be responsible for the signal at

i. m/z = 43 ....$CH_3CH_2CH_2^+$ or $C_3H_7^+$....

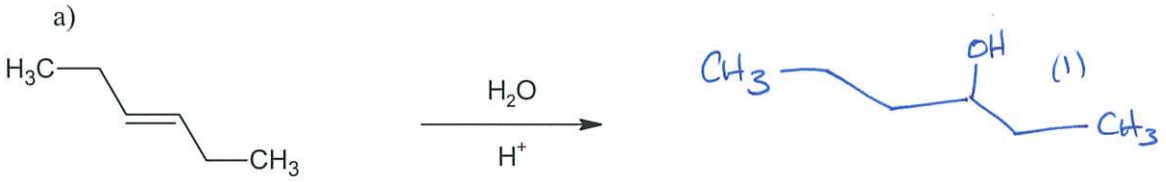
ii. m/z = 71 ....$CH_3CH_2CH_2CO^+$ or $C_4H_7O^+$....

- must have ⊕ charge but only penalise once.

Question 12

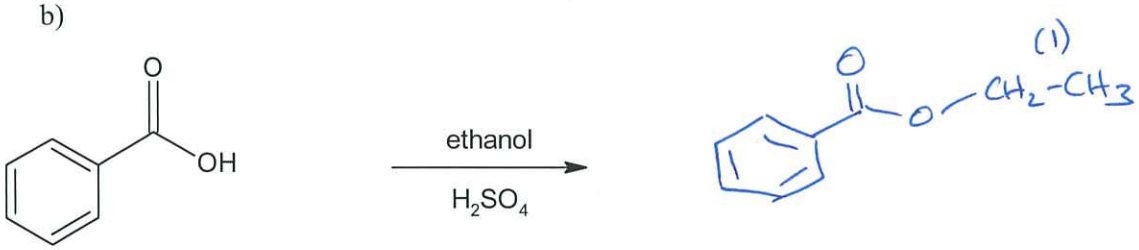
Draw the main organic products expected from the following reactions, and in each case state the type of reaction occurring. (2+2 marks)

a)



REACTION TYPE: ....ADDITION. (1)....

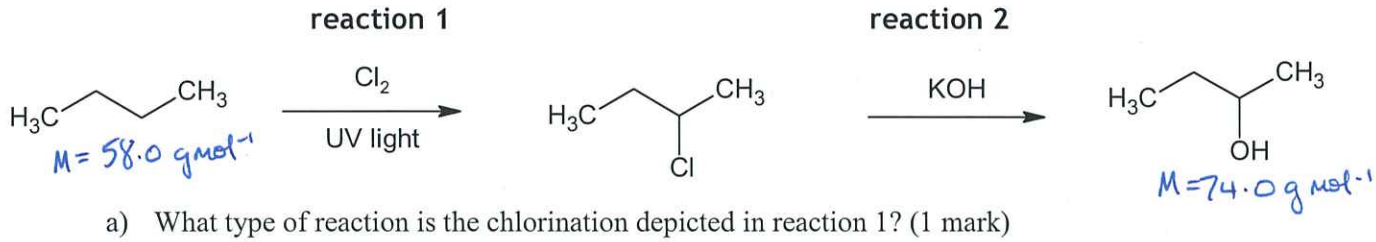
b)



REACTION TYPE: ....CONDENSATION. (1)....
(accept ESTERIFICATION)

Question 13

Consider the following reaction sequence for the preparation of butan-2-ol from butane.



a) What type of reaction is the chlorination depicted in reaction 1? (1 mark)

SUBSTITUTION (1)

b) In one case, starting with 350 g of butane, 320 g of butan-2-ol was obtained. What was the percentage yield? (2 marks)

theoretical yield : $M(\text{butan-2-ol}) = \frac{74.0}{58.0} \times 350 = 447\ g$ (1)

% yield $= \frac{320}{447} \times \frac{100}{1} = 71.7\ \%$. (1)

c) The overall process can be represented by the following balanced equation:

$$C_4H_{10} + Cl_2 + 2KOH \rightarrow C_4H_9OH + 2KCl + H_2O$$

58.0 71.0 56.1 74.0

Calculate the atom economy of this process. (2 marks)

(1)

$AE = \frac{74.0}{58.0 + 71.0 + 2(56.1)} \times \frac{100}{1} = 30.7\ \%$.

For idea : $AE = \frac{M(\text{desired product})}{M(\text{all reactants})}$ - (1)

End of Task – 56 marks available